# Surface Energies of a Chemically Disordered FCC Fe–25Cr–20Ni Alloy

MM309N – Computational Methods for Materials

Assignment 2 – Composition 1

Group 8

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26 October 2026

| | |
|---|---|
| **Alloy** | Fe–25Cr–20Ni (wt%), Type 310 based; 54.5/26.6/18.9 at.% Fe/Cr/Ni |
| **Surfaces, temperatures** | (111), (100), (110); 300, 500 and 700 K |
| **Potential** | Bonny–Castin–Terentyev (2013) Fe–Ni–Cr EAM |
| **Configurations** | Three random solid-solution configurations (seeds 11, 22, 33) |

**Abstract**

Surface energies $\gamma$ of the (111), (100) and (110) facets of a chemically disordered FCC Fe–25Cr–20Ni (wt%) alloy were computed at 300, 500 and 700 K with the Bonny–Castin–Terentyev EAM potential in LAMMPS. Three quenched-random configurations were used for every surface and temperature (27 slab calculations). The quantity reported is the potential-energy surface excess $\gamma = (\langle U_{\mathrm{slab}} \rangle - N \langle u_{\mathrm{bulk}} \rangle)/2A$, with the bulk reference computed at the same temperature; it contains no vibrational-entropy term. Slab thickness, in-plane size, vacuum, bulk-cell size and averaging length were tested on the (111) facet. The ordering $\gamma_{111} < \gamma_{100} < \gamma_{110}$ holds at all temperatures: $\gamma_{111} = 1.589 \rightarrow 1.562$, $\gamma_{100} = 1.678 \rightarrow 1.710$ and $\gamma_{110} = 1.860 \rightarrow 1.856$ J m$^{-2}$ from 300 to 700 K, with a configuration spread of 0.006–0.021 J m$^{-2}$ (at most 1.2%). In every configuration $\gamma_{111}$ decreases and $\gamma_{100}$ increases with temperature, while $\gamma_{110}$ is flat. The facet ordering and anisotropy resemble published EAM and DFT results for Ni, but the absolute values are 15–20% lower, and no direct reference for this alloy exists.

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# 1 Introduction and objectives

Surface energy is the energy needed to create a surface, and it depends on the exposed crystal plane because the number of broken bonds differs between planes. In a random alloy, the arrangement of Fe, Cr and Ni atoms at the surface also changes the energy from one configuration to another. This report determines $\gamma_{111}$, $\gamma_{100}$ and $\gamma_{110}$ of Type 310-based Fe–25Cr–20Ni at 300, 500 and 700 K, and tests whether the values are converged and reproducible. The research questions are:

1. What is the ordering and magnitude of $\gamma$ for the three facets?
2. How large is the variation between independent chemical configurations?
3. Does $\gamma$ change systematically between 300, 500 and 700 K?
4. Are the values stable with respect to slab thickness, in-plane size, vacuum, bulk reference and sampling?

## 1.1 Predictions made before running the calculations

- **Facet ordering:** from the broken-bond picture, $\gamma_{111} < \gamma_{100} < \gamma_{110}$, since (111) is the most densely packed plane.
- **Temperature:** $\gamma$ is expected to decrease with temperature because thermal expansion and softening weaken the bonds that are cut.
- **Chemical disorder:** configurations will differ measurably, and the open (110) facet will show a larger spread than (111).

# 2 Interatomic potential

The calculations use the EAM-type Fe–Ni–Cr potential of Bonny, Castin and Terentyev [1], obtained from the NIST repository [2]. It was developed for ageing and irradiation studies in stainless steels, with attention to point-defect behaviour near Fe–10Ni–20Cr. It was not fitted to surface energies, so its suitability for surfaces is an assumption that the checks in Sec. **??** only partly test.

**Table 1:** Potential provenance and implementation.

| Item | Details |
|---|---|
| Formalism | Embedded atom method, `pair_style eam/alloy` |
| File | `FeNiCr_Bonny_2013_ptDef.eam.alloy`; NIST entry `2013-Bonny-G-Castin-N-Terentyev-D-Fe-Ni-Cr`, retrieved October 2026 |
| Mapping | `pair_coeff * * <file> Fe Cr Ni`: type 1 = Fe, 2 = Cr, 3 = Ni (matched to the file by element name) |
| Interactions | Fe, Cr, Ni, with all cross terms in the file |
| Composition range | Developed near Fe–10Ni–20Cr (at.%); this alloy (Fe–26.6Cr–18.9Ni at.%) has more Cr |
| Benchmarks in the paper | Point-defect formation and migration, vacancy–solute binding, interstitial clustering, diffusion |
| Limitations | No magnetism; parametrised for bulk/defect energetics, not surfaces; no segregation validation |

# 3 Computational methodology

## 3.1 Composition and configurations

The alloy is specified in weight percent, which converts to Fe 54.5, Cr 26.6, Ni 18.9 at.%. Each configuration is a quenched random substitutional FCC solid solution: Fe, Cr and Ni labels are assigned in fixed integer counts, rounded by the largest-remainder rule so they sum exactly to $N$ (for $N = 512$: Fe 279, Cr 136, Ni 97), shuffled with a seeded NumPy generator (`default_rng`) and placed on the sites of a perfect FCC lattice. Seeds 11, 22 and 33 give configurations C1, C2 and C3, which are independent samples of the disordered state. Because the shuffle depends on $N$, a given seed gives a different arrangement in the bulk cell than in a slab. Only atomic positions relax; species never swap, so there is no Monte Carlo equilibration, no short-range order and no surface segregation. This is a limitation of the model.

**Table 2:** Atom counts of the configurations (the same for all three seeds).

| Cell | $N$ | Fe / Cr / Ni |
|---|---|---|
| (111) slab | 512 | 279 / 136 / 97 |
| (100) slab | 576 | 314 / 153 / 109 |
| (110) slab | 576 | 314 / 153 / 109 |

## 3.2 Slab construction

Slabs were built with ASE (`fcc111`, `fcc100`, `fcc110`, orthogonal cells) using the lattice parameter $a(T)$ of the bulk NPT runs at the same temperature. Boundaries are periodic in all directions, with a 20 Å empty gap along the surface normal. In the ideal lattice the two exposed surfaces are equivalent and the slab is symmetric about its midplane; with random chemistry they are statistically equivalent. The in-plane box is fixed at the bulk $a(T)$. The (111) model was chosen by the convergence study of Sec. 4; for (100) and (110) the thickness was matched in Å and the in-plane size chosen from the printed cell dimensions.

**Table 3:** Production slab models (lengths at 300 K).

| Facet | Layers | Thickness (Å) | In-plane cell | Size (Å) | Atoms |
|---|---|---|---|---|---|
| (111) | 8 | 14.4 | $8 \times 8$ | $20.2 \times 17.5$ | 512 |
| (100) | 9 | 14.3 | $8 \times 8$ | $20.2 \times 20.2$ | 576 |
| (110) | 12 | 13.9 | $8 \times 6$ | $28.6 \times 15.2$ | 576 |

### 3.3 Simulation protocol

**Table 4:** Simulation parameters.

| Parameter | Value |
|---|---|
| Software | LAMMPS (version not recorded); ASE, NumPy, pandas, matplotlib |
| Integrator, time step | Velocity-Verlet, 2 fs |
| 0 K bulk | Conjugate-gradient minimisation (tolerances $10^{-12}$–$10^{-14}$) with `box/relax aniso 0` |
| Finite-$T$ bulk | NPT, Nosé–Hoover, 0 bar, $T_{\mathrm{damp}} = 0.2$ ps, $P_{\mathrm{damp}} = 2.0$ ps; 40 ps equilibration + 100 ps averaging (500 atoms) |
| Slab | Minimisation (tolerance $10^{-10}$, 5000 iterations, 50000 evaluations), then NVT, $T_{\mathrm{damp}} = 0.2$ ps; 20 ps equilibration + 50 ps averaging |
| Averaging | Slab: potential energy every 10 steps (running average). Bulk: 50 samples over 100 ps |
| Initial velocities | Gaussian at $T$, random seed equal to the configuration seed |

### 3.4 Bulk reference

The bulk reference is the mean potential energy per atom of three random 500-atom cells (seeds 11, 22, 33), relaxed at 0 K or averaged in NPT at each temperature. The same value (the mean of the three cells) is used for all slabs at a given temperature. At 256 atoms the 0 K energy had the same mean ($-4.1201$ eV/atom) but a spread of 6.7 meV/atom, six times larger than at 500 atoms (1.2 meV/atom); this would have changed $\gamma$ by about 0.3 J m$^{-2}$ for a slab of that size, so 500-atom cells were adopted.

**Table 5:** Bulk reference (mean of three 500-atom cells ± SD).

| $T$ (K) | $a$ (Å) | $u_{\mathrm{bulk}}$ (eV/atom) |
|---|---|---|
| 0 | 3.5508 | $-4.12008 \pm 0.00119$ |
| 300 | 3.5745 | $-4.07956 \pm 0.0013$ |
| 500 | 3.5932 | $-4.04909 \pm 0.0011$ |
| 700 | 3.6142 | $-4.01573 \pm 0.0013$ |

An error of 1 meV/atom in $u_{\mathrm{bulk}}$ shifts $\gamma$ of a 512–576 atom slab by about 0.015–0.02 J m$^{-2}$. This offset is common to all slabs at one temperature, so it affects absolute values more than trends.

### 3.5 Definition of surface energy

$$\gamma_{hkl}^{(c)}(T) = \frac{\langle U_{\mathrm{slab,hkl}}^{(c)}(T)\rangle - N\,\langle u_{\mathrm{bulk}}(T)\rangle}{2A_{hkl}} \times 16.021766, \tag{1}$$

where $c$ labels the configuration, $N$ is the number of atoms, $A = L_x L_y$ is the area of one face, the factor 2 accounts for the two faces, and the constant converts eV/Å$^2$ to J m$^{-2}$. $\langle U_{\mathrm{slab}}\rangle$ is the time average over the production window. Eq. (1) is a potential-energy surface excess. It is *not* a surface free energy, because the vibrational-entropy contribution is omitted. For each surface and temperature the mean over the three configurations and the sample standard deviation ($n-1$ normalisation) are reported, together with the standard error SE $= s/\sqrt{n}$.

## 4 Convergence and validation

The convergence study was run on the (111) facet at 0 K with seeds 11, 22, 33 and $u_{\mathrm{bulk}} = -4.12008$ eV/atom (the tests were first run with five seeds, but only seeds 11, 22, 33 are reported). The decision criterion was no systematic trend with the varied parameter, with $|\Delta\gamma|$ not larger than the configuration scatter at that size (about 0.02–0.08 J m$^{-2}$).

**Table 6:** Convergence tests, (111), Comp 1, 0 K (mean over seeds 11, 22, 33). Baseline: 8 layers, $4 \times 4$, 20 Å.

| Test | Layers / atoms | In-plane | Vacuum (Å) | $\gamma$ | $\lvert\Delta\gamma\rvert$ | Note |
|---|---|---|---|---|---|---|
| Baseline | 8 / 128 | $4 \times 4$ | 20 | 1.614 | – | SD 0.016 |
| Thickness | 6 / 96 | $4 \times 4$ | 20 | 1.631 | 0.017 | within scatter |
| Thickness | 10 / 160 | $4 \times 4$ | 20 | 1.659 | 0.045 | SD 0.076; no trend |
| Thickness | 12 / 192 | $4 \times 4$ | 20 | 1.620 | 0.006 | SD 0.078; no trend |
| In-plane | 8 / 288 | $6 \times 6$ | 20 | 1.636 | 0.022 | SD 0.072 |
| In-plane | 8 / 512 | $8 \times 8$ | 20 | 1.612 | 0.002 | SD 0.006; adopted |
| Vacuum | 8 / 128 | $4 \times 4$ | 10–30 | 1.614 | 0.000 | identical |
| Vacuum | 8 / 128 | $4 \times 4$ | 4 | 1.576$^{\dagger}$ | 0.020$^{\dagger}$ | ∼1.3% lower |

$^{\dagger}$ Seed 11 only, compared with seed 11 at 20 Å (1.596), using the provisional $u_{\mathrm{bulk}} = -4.1201$ eV/atom.

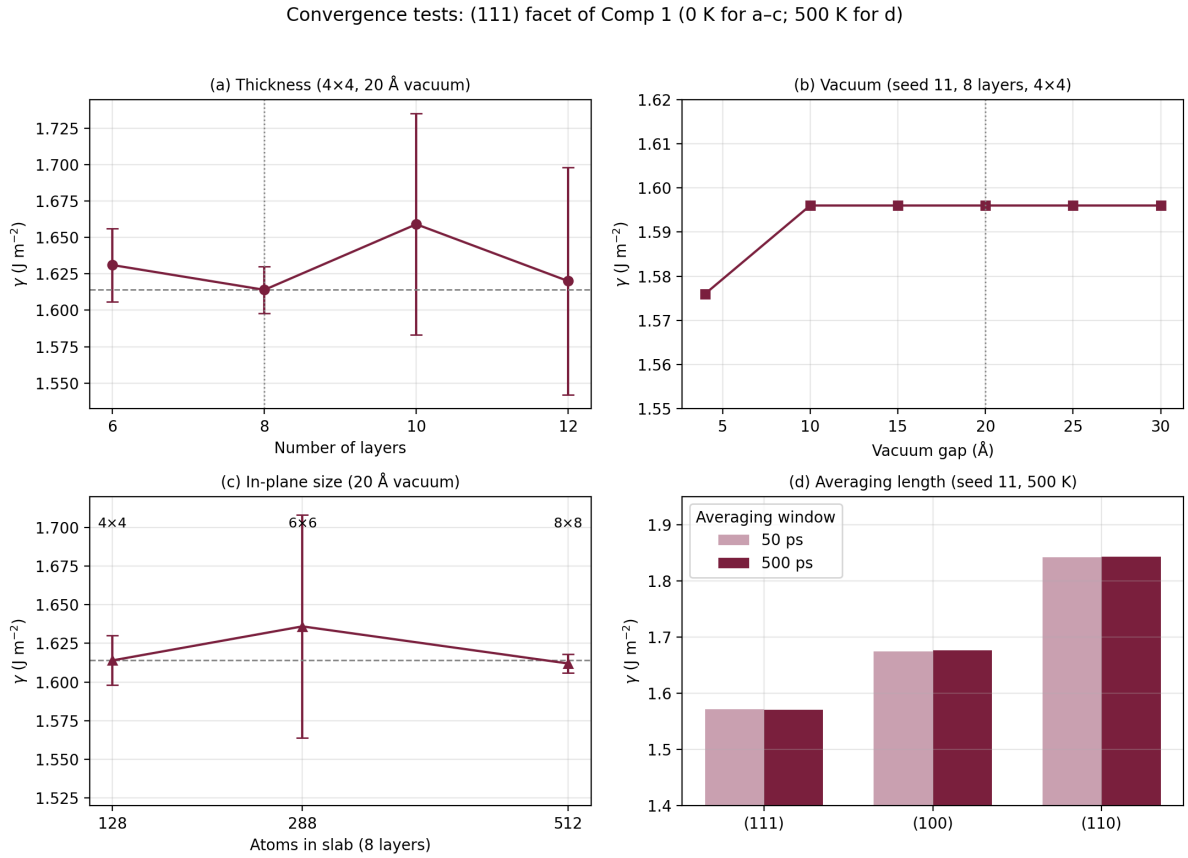



**Figure 1:** Convergence tests on the (111) facet: (a) thickness, (b) vacuum (seed 11), (c) in-plane size, (d) averaging length (seed 11, 500 K). Error bars in (a) and (c) are the SD over three configurations.

**Thickness and size.** $\gamma$ shows no trend from 6 to 12 layers, and the differences are within the scatter, so 8 layers (about 14.4 Å) is adopted with a margin; 6 layers would also pass. In-

plane size has no effect on the mean, and the scatter is smallest at $8 \times 8$ (SD 0.006), which is adopted. The larger scatter at $6 \times 6$ and at 10 and 12 layers comes from three samples and is not interpreted.

**Vacuum.** $\gamma$ is identical to every printed digit for 10 to 30 Å, because all of these gaps exceed the potential cutoff. The 4 Å run lowers $\gamma$ by 0.020 J m$^{-2}$ (1.3%), showing the test can detect interaction between periodic images. 20 Å is used.

**Averaging length.** For seed 11 at 500 K, each facet was rerun with 100 ps equilibration and 500 ps averaging, ten times longer than the production 50 ps. $\gamma$ changed by at most 0.001 J m$^{-2}$ ($< 0.1\%$), far below the configuration scatter (0.009–0.020), so the 50 ps window is adequate (Table 7).

**Table 7:** Averaging-length test, seed 11, 500 K ($u_{\mathrm{bulk}} = -4.04909$ eV/atom).

| Facet | $\gamma$, 50 ps | $\gamma$, 500 ps | Difference |
|---|---|---|---|
| (111) | 1.572 | 1.571 | −0.001 |
| (100) | 1.675 | 1.676 | +0.001 |
| (110) | 1.842 | 1.843 | +0.001 |

**Bulk and lattice checks.** The 0 K lattice parameter is 3.551 Å, about 1% below typical experimental values for austenitic stainless steel (≈3.58–3.59 Å). Between 300 and 700 K the mean linear expansion is about $2.8 \times 10^{-5}$ K$^{-1}$, larger than the roughly $1.7 \times 10^{-5}$ K$^{-1}$ usually quoted for austenitic steels (a typical handbook value, not independently cited here), so the potential overestimates thermal expansion; this is carried into the slabs through $a(T)$.

**Internal consistency.** Recomputing the surface area implied by every $\gamma$ value gives the same area for all three seeds at each surface and temperature (for example 354.1, 357.8 and 362.0 Å$^2$ for (111) at 300, 500 and 700 K), matching the lattice parameters and printed cell dimensions.

**Not tested.** The time step (2 fs) was not varied. Convergence was tested only on (111) at 0 K, and the settings were transferred to (100) and (110). No structural analysis (for example common-neighbour analysis) was run, so the FCC structure at 700 K is assumed from the smooth trends in $a$ and $u$.

## 5 Results

**Table 8:** Surface energies (J m$^{-2}$) of the individual configurations, mean, SD and SE ($n = 3$).

| Facet | $T$ (K) | C1 | C2 | C3 | Mean | SD | SE |
|---|---|---|---|---|---|---|---|
| (111) | 300 | 1.587 | 1.596 | 1.585 | 1.589 | 0.006 | 0.003 |
| | 500 | 1.572 | 1.585 | 1.569 | 1.575 | 0.009 | 0.005 |
| | 700 | 1.549 | 1.578 | 1.559 | 1.562 | 0.015 | 0.009 |
| (100) | 300 | 1.665 | 1.699 | 1.669 | 1.678 | 0.018 | 0.010 |
| | 500 | 1.675 | 1.712 | 1.682 | 1.690 | 0.020 | 0.012 |
| | 700 | 1.696 | 1.734 | 1.701 | 1.710 | 0.020 | 0.012 |
| (110) | 300 | 1.848 | 1.880 | 1.852 | 1.860 | 0.018 | 0.010 |
| | 500 | 1.842 | 1.876 | 1.846 | 1.855 | 0.018 | 0.010 |
| | 700 | 1.841 | 1.880 | 1.846 | 1.856 | 0.021 | 0.012 |

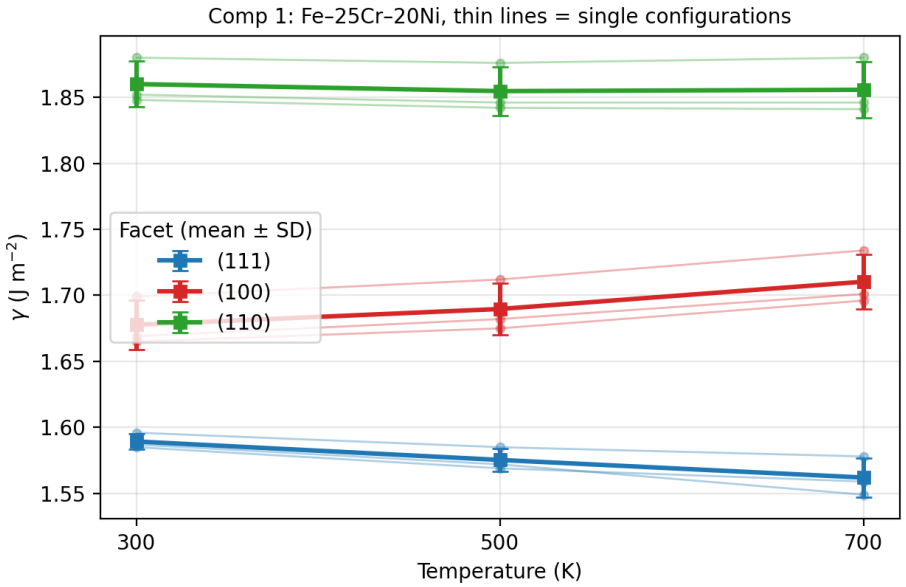



**Figure 2:** Surface energy versus temperature. Thin lines are single configurations; thick lines are the mean with SD error bars.

# 6 Discussion

## 6.1 Facet ordering

The ordering $\gamma_{111} < \gamma_{100} < \gamma_{110}$ holds at every temperature and in every individual configuration, as predicted from the broken-bond picture. At 300 K the gaps are 0.089 J m$^{-2}$ between (111) and (100) and 0.182 between (100) and (110); at 700 K they are 0.148 and 0.146. All gaps are several times larger than the configuration SD (0.006–0.021), so the ordering is statistically clear.

## 6.2 Temperature dependence

From 300 to 700 K, $\gamma_{111}$ falls by 0.027 J m$^{-2}$ (changes of −0.038, −0.018, −0.026 in the three configurations), $\gamma_{100}$ rises by 0.032 (+0.032, +0.035, +0.031) and $\gamma_{110}$ is flat (−0.007, 0.000, −0.006). Because the same three configurations are used at every temperature, each keeps nearly the same offset from the mean, so these changes are more consistent than the SD alone suggests. The predicted decrease with temperature is seen only for (111); (110) shows no change and (100) rises, contrary to the prediction. The cause was not isolated. Since only potential energies are used, the missing vibrational entropy of the surface could change both the size and the sign of the trend, and no statement about the surface free energy is made. The potential also overestimates thermal expansion.

## 6.3 Chemical disorder and sampling

The configuration SD is 0.006–0.015 J m$^{-2}$ for (111) and 0.018–0.021 for (100) and (110), at most 1.2% of $\gamma$. The (110) spread is larger than the (111) spread, as predicted, but (100) is as large as (110). Three configurations are enough to separate the facets (gaps of 0.09–0.18 J m$^{-2}$) but not to resolve differences below about 0.02–0.03 J m$^{-2}$, and each SD from $n = 3$ is itself uncertain. The tenfold-longer averaging test shows that the 50 ps window is not a significant error source.

### 6.4 Comparison with literature

No published surface energies exist for this alloy with this potential. The closest comparison is pure Ni, a fcc metal with the same facets (Table 9); the values are taken from a review table [3] and are not from the same potential, so they test plausibility rather than accuracy.

**Table 9:** Comparison with pure Ni ($\mathrm{J\,m^{-2}}$). Ni values from Ref. [3].

| | (111) | (100) | (110) |
|---|---|---|---|
| Ni, EAM [3] | 1.949 | 2.041 | 2.231 |
| Ni, DFT [3] | 1.92 | 2.21 | 2.29 |
| Ni, experiment (facet average) [3] | | 2.380 | |
| This work, 300 K | 1.589 | 1.678 | 1.860 |

The facet ordering is the same as for Ni, and the anisotropy is similar: $\gamma_{100}/\gamma_{111} = 1.056$ and $\gamma_{110}/\gamma_{111} = 1.171$ here, against 1.047 and 1.145 for Ni EAM. The absolute values are 15–20% lower than Ni EAM and DFT. This is plausible for several reasons: the alloy contains only about 19 at.% Ni, the potential was not fitted to surface energies, EAM potentials are known to underestimate fcc surface energies [4], and the slabs here are relaxed and at finite temperature. The experimental value is an average over many facets and cannot be compared with a single orientation. The absolute values should therefore be treated as model-dependent.

### 6.5 Limitations

- **Potential:** not fitted to surface energies, no magnetism, more Cr than the target composition, and thermal expansion overestimated.
- **Energy, not free energy:** no vibrational entropy.
- **Sizes:** convergence tested on (111) at 0 K only; (100) and (110) matched in thickness; the time step was not tested.
- **Sampling:** $n = 3$; averaging checked for one configuration per facet at 500 K only.
- **Bulk reference:** a three-cell mean with SD about 1–1.3 meV/atom, giving a common offset of about 0.02 $\mathrm{J\,m^{-2}}$.
- **No segregation or short-range order:** species are fixed on random sites.
- **No structural analysis** of the 700 K runs.

## 7 Conclusions

1. $\gamma_{111} < \gamma_{100} < \gamma_{110}$ at all temperatures, e.g. $1.589 \pm 0.006$, $1.678 \pm 0.018$ and $1.860 \pm 0.018$ $\mathrm{J\,m^{-2}}$ at 300 K (mean $\pm$ SD, $n = 3$), consistent with the broken-bond picture.
2. From 300 to 700 K, $\gamma_{111}$ decreases by 0.027 $\mathrm{J\,m^{-2}}$, $\gamma_{100}$ increases by 0.032 and $\gamma_{110}$ is unchanged, with the same sign in every configuration. Only (111) agrees with the predicted decrease.
3. The configuration spread is 0.006–0.021 $\mathrm{J\,m^{-2}}$ ($\leq$ 1.2%), small compared with the facet differences.
4. The (111) model (8 layers, $8 \times 8$, 20 Å vacuum) is converged within the configuration scatter at 0 K, and a tenfold longer averaging window changes $\gamma$ by $\leq 0.001$ $\mathrm{J\,m^{-2}}$.

5. The facet ordering and anisotropy resemble pure-Ni results, but the absolute values are 15–20% lower and are model-dependent; the main limitations are the potential, the potential-energy definition and $n = 3$.

## Reproducibility

**Table 10:** Files and their roles.

| File | Role |
|---|---|
| `make_bulk.py`, `make_bulk5.py` | Build random 256- and 500-atom bulk cells (seeds 11, 22, 33) |
| `bulk_0K.in` | Relax bulk cells at 0 K; prints $a$ and energy per atom |
| `bulk_NPT.in`, `avg_npt.py` | NPT at 300/500/700 K; average $a(T)$ and $u_{\mathrm{bulk}}(T)$ |
| `make_slab.py` | Build random-alloy slabs for a chosen facet, size, layers, vacuum, seed and $a$ |
| `slab_0K.in` | Relax slabs at 0 K (convergence tests) and compute $\gamma$ |
| `run_batch.py`, `run_vacuum.py`, `run_inplane.py` | Loop over thickness, vacuum and in-plane tests |
| `recompute3.py`, `recompute_gamma.py` | Keep seeds 11/22/33; recompute $\gamma$ with the final $u_{\mathrm{bulk}}$ |
| `slab_NVT.in`, `run_slabs.py` | Production NVT runs and $\gamma$ for the 27 cases; writes `gamma_production.csv` |
| `slab_NVT_long.in`, `compare_long.py` | 500 ps averaging test |
| `potentials/` | `FeNiCr_Bonny_2013_ptDef.eam.alloy` |

## A Statistical definitions

For $n$ configuration-level values $\gamma_i$, the sample mean is $\bar{\gamma} = \frac{1}{n}\sum_i \gamma_i$, the sample standard deviation is $s = \sqrt{\frac{1}{n-1}\sum_i(\gamma_i - \bar{\gamma})^2}$, and the standard error is $s/\sqrt{n}$.

## References

[1] G. Bonny, N. Castin, D. Terentyev, "Interatomic potential for studying ageing under irradiation in stainless steels: the FeNiCr model alloy," *Modelling Simul. Mater. Sci. Eng.* **21**, 085004 (2013). doi:10.1088/0965-0393/21/8/085004.

[2] NIST Interatomic Potentials Repository, Fe–Ni–Cr potential of Bonny, Castin and Terentyev (2013). `https://www.ctcms.nist.gov/potentials/entry/2013--Bonny-G-Castin-N-Terentyev-D--Fe-Ni-Cr/`. Accessed 26 October 2026.

[3] "Polarizable Potentials for Metals: The Density Readjusting Embedded Atom Method (DR-EAM)," arXiv:1904.00263 (surface-energy table, Table IV).

[4] "Applications of a tight-binding total energy method for transition and noble metals: Elastic Constants, Vacancies, and Surfaces of Monatomic Metals," arXiv:mtrl-th/9602003.